

Combinatorial Mathematics



Chapter 4 Number Theory and Cryptography

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- ⊕ 4.1 Divisibility and Modular Arithmetic
 - ⊕ 4.2 Integer Representations and Algorithms
 - ⊕ 4.3 Primes and Greatest Common Divisors
 - ⊕ 4.4 Solving Congruences
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 - ⊕ 4.6 Cryptography



4.1 The integers and division

※探討一些 Number Theory 的基本觀念

Def 1. a, b : integers, $a \neq 0$.

a divides b (denote $a \mid b$) if $\exists c \in \mathbb{Z}$, $b = ac$.

(a : a **factor** of b , b : a **multiple** of a)

($a \nmid b$ if a does not divide b)

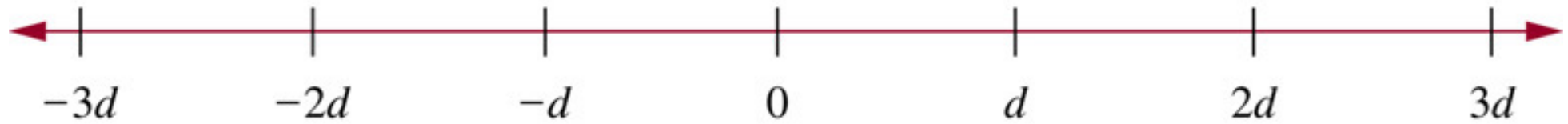


FIGURE 1 Integers Divisible by the Positive Integer d .

- **Example 1** : Determine whether $3 \mid 7$ and whether $3 \mid 12$

■ **Sol** :

$3 \nmid 7$ ，因為 $7/3$ 不是整數。但是， $3 \mid 12$ ，因為 $12/3 = 4$ 。

- **Example 2** : let n and d be positive integers. How many positive integers not exceeding n are divisible by d ?

■ **Sol** :

所有能被 d 整除的正整數能以 dk 表示，其中 k 為一個正整數。因為要找出小於 n 的正整數，即 $0 < dk \leq n$ ，也可表為 $0 < k \leq n/d$ 。所以一共有 $\lfloor n/d \rfloor$ 個小於 n 的正整數能被 d 整除。

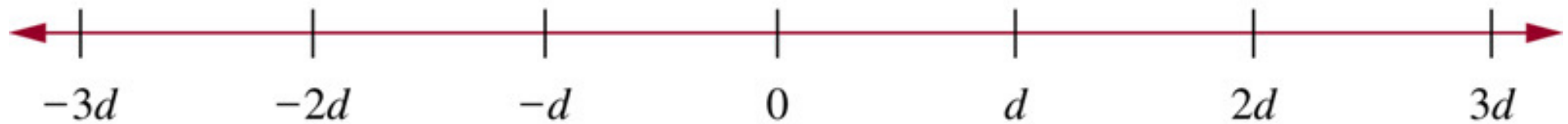


FIGURE 1 Integers Divisible by the Positive Integer d .

■ **Theorem 1 :** Let a , b , and c be integer, where $a \neq 0$

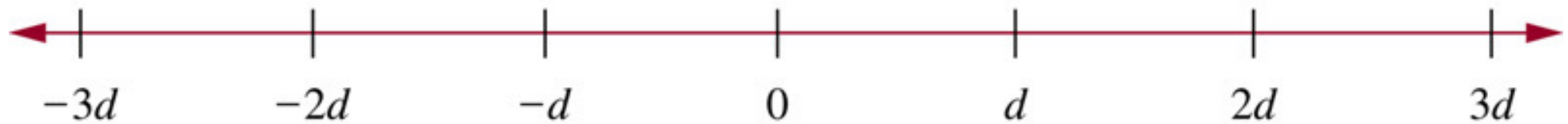
- (1) if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$
- (2) if $a \mid b$, then $a \mid bc$ for all integers c
- (3) if $a \mid b$ and $b \mid c$, then $a \mid c$.

Corollary 1. if $a, b, c \in \mathbb{Z}$ and $a \mid b$, $a \mid c$.

then $a \mid mb + nc$ whenever $m, n \in \mathbb{Z}$

Def 2. In the equality $a = dq + r$ with $0 \leq r < d$, d is called the **divisor** (除數), a is called the **dividend** (被除數), q is called the **quotient** (商數), and r is called the **remainder** (餘數).

$$\Rightarrow q = a \operatorname{div} d, \quad r = a \operatorname{mod} d$$



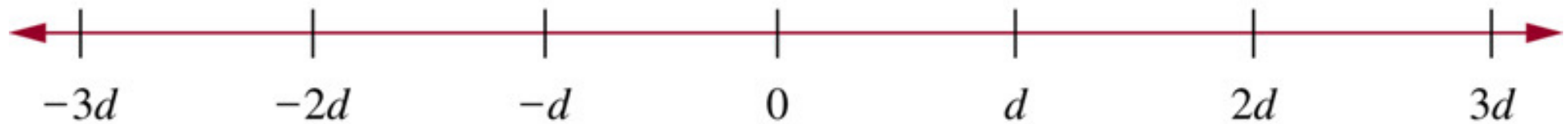
■ **Def 2.** : $q = a \text{ div } d$, $r = a \text{ mod } d$

■ **Example 3** : What are the **quotient** (商數) and **remainder** (餘數) when 101 is divided by 11 ?

■ **Sol** : 我們有 $101 = 11 \cdot 9 + 2$ 。

所以，當101除以11時，商數為 $9 = 101 \text{ div } 11$ 。

而餘數為 $2 = 101 \text{ mod } 11$ 。



■ **Def 2.** : $q = a \text{ div } d$, $r = a \text{ mod } d$

■ **Example 4** : What are the **quotient** (商數) and **remainder** (餘數) when -11 is divided by 3 ?

■ **Sol** : $\because -11 = 3 \cdot (-4) + 1$

所以，當 -11 除以 3 時，商數為 $-4 = -11 \text{ div } 3$ 。

而餘數為 $1 = -11 \text{ mod } 3$ 。

必須注意的是，餘數不能是負數。所以餘數不會是 -2 ，雖然 $-11 = 3(-3) - 2$ 。

因為， $r = -2$ 並不滿足 $0 \leq r < 3$ 。

Modular Arithmetic

Def 3. If $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$, then
 a is **congruent (同餘) to b modulo m** if $m \mid (a-b)$.
(模 m 時 a 同餘於 b)

(**Thm 3.** denote $a \equiv b \pmod{m}$ iff $a \bmod m = b \bmod m$).

Thm 4. Let $m \in \mathbb{Z}^+$, $a, b \in \mathbb{Z}$.

$$a \equiv b \pmod{m} \text{ iff } \exists k \in \mathbb{Z}, \text{ s.t. } a = b + km.$$

Thm 5. Let $m \in \mathbb{Z}^+$, $a, b \in \mathbb{Z}$.

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$,
then $a + c \equiv b + d \pmod{m}$
and $ac \equiv bd \pmod{m}$.

Corollary 2.

$$(a+b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$
$$ab \bmod m = ((a \bmod m)(b \bmod m)) \bmod m.$$

Def 3. If $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$, then

a is **congruent (同餘) to b modulo m** if $m \mid (a-b)$.

(**Thm 3.** denote $a \equiv b \pmod{m}$ iff $a \bmod m = b \bmod m$).

- **Example 5 :** Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.

- **Sol :**

因為6整除 $17 - 5 = 12$ ，可知在模6時17同餘於5。

而6無法整除 $24 - 14 = 10$ ，所以在模6時24不同餘於14。

Modular Arithmetic

Def 3. If $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$, then
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and $ac \equiv bd \pmod{m}$.

Corollary 2.

$$(a+b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$
$$ab \bmod m = ((a \bmod m)(b \bmod m)) \bmod m.$$



More on Congruences

Theorem 4: Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof:

- If $a \equiv b \pmod{m}$, then (by the definition of congruence) $m \mid a - b$. Hence, there is an integer k such that

$$a - b = km \text{ and equivalently } a = b + km.$$

- Conversely, if there is an integer k such that $a = b + km$, then $km = a - b$. Hence, $m \mid a - b$ and $a \equiv b \pmod{m}$.

Modular Arithmetic

Def 3. If $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$, then
 a is **congruent (同餘)** to b modulo m if $m \mid (a-b)$.
(模 m 時 a 同餘於 b)

(**Thm 3.** denote $a \equiv b \pmod{m}$ iff $a \bmod m = b \bmod m$).

Thm 4. Let $m \in \mathbb{Z}^+$, $a, b \in \mathbb{Z}$.

$$a \equiv b \pmod{m} \text{ iff } \exists k \in \mathbb{Z}, \text{ s.t. } a = b + km.$$

- Relation between " $x \equiv b \pmod{m}$ " and " $x = b \bmod m$ "

Example : The equation $x \equiv 16 \pmod{10}$

has solutions $x = \dots, -24, -14, -4, 6, 16, 26, 36, 46, \dots$

equal to $x \equiv -24 \equiv -14 \equiv -4 \equiv 6 \equiv 16 \equiv 26 \equiv 36 \equiv 46 \pmod{10}$.

$a \bmod m$ denotes **the smallest positive number** x such that $x \equiv a \pmod{m}$.

So $6 = 16 \bmod 10$ and $6 = -24 \bmod 10$ etc...



$$0 < a \bmod m < m$$

Modular Arithmetic

Def 3. If $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$, then
 a is **congruent (同餘)** to b modulo m if $m \mid (a-b)$.
(模 m 時 a 同餘於 b)

(**Thm 3.** denote $a \equiv b \pmod{m}$ iff $a \bmod m = b \bmod m$).

Thm 4. Let $m \in \mathbb{Z}^+$, $a, b \in \mathbb{Z}$.

$$a \equiv b \pmod{m} \text{ iff } \exists k \in \mathbb{Z}, \text{ s.t. } a = b + km.$$

- Relation between " $x \equiv b \pmod{m}$ " and " $x = b \bmod m$ "

Example : The standard representatives for all possible numbers modulo 10 are given by "0, 1, 2, 3, 4, 5, 6, 7, 8, 9"
e.g. $3 \equiv 13 \equiv 23 \pmod{10}$,
we would take the smallest positive such number which is 3.

Modular Arithmetic

Def 3. If $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$, then
 a is **congruent (同餘) to b modulo m** if $m \mid (a-b)$.
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Thm 4. Let $m \in \mathbb{Z}^+$, $a, b \in \mathbb{Z}$.
 $a \equiv b \pmod{m}$ iff $\exists k \in \mathbb{Z}$, s.t. $a = b + km$.

Thm 5. Let $m \in \mathbb{Z}^+$, $a, b \in \mathbb{Z}$.
If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$,
then $a + c \equiv b + d \pmod{m}$(sum rule)
and $ac \equiv bd \pmod{m}$(multiplication rule)

Coro $\begin{cases} a \equiv b \pmod{m} \\ c \equiv d \pmod{m} \end{cases} \longrightarrow \begin{cases} a + c \equiv b + d \pmod{m} \\ a \times c \equiv b \times d \pmod{m} \end{cases}$
 $ab \pmod{m} = ((a \pmod{m})(b \pmod{m})) \pmod{m}$.

Congruences of Sums and Products

Theorem 5: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$

Proof:

- Because $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, by Theorem 4 there are integers s and t with $b = a + sm$ and $d = c + tm$.
- Therefore,
 - $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and
 - $bd = (a + sm)(c + tm) = ac + m(at + cs + stm)$.
- Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Example 6 : Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from Theorem 5 that

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$$

$$77 = 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod{5}$$

Modular Arithmetic

Def 3. If $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$, then
 a is **congruent (同餘) to b modulo m** if $m \mid (a-b)$.
(模 m 時 a 同餘於 b)
(**Thm 3.** denote $a \equiv b \pmod{m}$).

Thm 4. Let $m \in \mathbb{Z}^+$, $a, b \in \mathbb{Z}$.
 $a \equiv b \pmod{m}$ iff $\exists k \in \mathbb{Z}$, s.t. $a = b + km$.

Thm 5. Let $m \in \mathbb{Z}^+$, $a, b \in \mathbb{Z}$.

$$\begin{array}{c} \left\{ \begin{array}{l} a \equiv b \pmod{m} \\ c \equiv d \pmod{m} \end{array} \right. \xrightarrow{\text{and } c \equiv d \pmod{m}} \left\{ \begin{array}{l} a + c \equiv b + d \pmod{m} \\ a \times c \equiv b \times d \pmod{m} \end{array} \right. \\ \text{and } ac \equiv bd \pmod{m}. \end{array}$$

Corollary 2.

$$\begin{aligned} (a+b) \bmod m &= ((a \bmod m) + (b \bmod m)) \bmod m \\ ab \bmod m &= ((a \bmod m)(b \bmod m)) \bmod m. \end{aligned}$$

■ **Exercise** : Find each of these values

23.(a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$

23.(b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$

24.(a) $(99^2 \bmod 32)^3 \bmod 15$

■ **Exercise** : Find each of these values

23.(a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$

$$\equiv (-133 + 261) \bmod 23 \equiv (128) \bmod 23$$

$$\equiv (5 \bmod 23) + (8 \bmod 23) = 13 \bmod 23$$

23.(b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$

24.(a) $(99^2 \bmod 32)^3 \bmod 15$

■ **Exercise** : Find each of these values

23.(a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$

23.(b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$

$$\equiv ((20 \bmod 23) \cdot (21 \bmod 23)) \bmod 23$$

$$\equiv (20 \cdot 21) \bmod 23$$

$$\equiv 6 \bmod 23$$

24.(a) $(99^2 \bmod 32)^3 \bmod 15$

■ **Exercise** : Find each of these values

23.(a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$

23.(b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$

24.(a) $(99^2 \bmod 32)^3 \bmod 15$

$$\equiv (3 \cdot 3 \bmod 32)^3 \bmod 15 \equiv 729 \bmod 15$$

$$\equiv ((720 \bmod 15) + (9 \bmod 15)) = 9 \pmod{15}$$

Arithmetic Modulo m

Definitions: Let $\mathbf{Z}_m$ be the set of nonnegative integers less than m : $\{0, 1, \dots, m-1\}$

- The operation $+_m$ is defined as $a +_m b = (a + b) \bmod m$.
(*addition modulo m*)
- The operation $\cdot_m$ is defined as $a \cdot_m b = (a \cdot b) \bmod m$.
(*multiplication modulo m*)
- Using these operations is said to be doing *arithmetic modulo m* .

Example: Find $7 +_{11} 9$ and $7 \cdot_{11} 9$.

Solution: Using the definitions above:

- $7 +_{11} 9 = (7 + 9) \bmod 11 = 16 \bmod 11 = 5$
- $7 \cdot_{11} 9 = (7 \cdot 9) \bmod 11 = 63 \bmod 11 = 8$

Arithmetic Modulo m

The operations $+_m$ and $\cdot_m$ satisfy many of the same properties as ordinary addition and multiplication.

- *Closure*: If a and b belong to $\mathbf{Z}_m$, then $a +_m b$ and $a \cdot_m b$ belong to $\mathbf{Z}_m$.
- *Associativity*: If a , b , and c belong to $\mathbf{Z}_m$, then $(a +_m b) +_m c = a +_m (b +_m c)$ and $(a \cdot_m b) \cdot_m c = a \cdot_m (b \cdot_m c)$.
- *Commutativity*: If a and b belong to $\mathbf{Z}_m$, then $a +_m b = b +_m a$ and $a \cdot_m b = b \cdot_m a$.
- *Identity elements*: The elements 0 and 1 are identity elements for addition and multiplication modulo m , respectively.
 - If a belongs to $\mathbf{Z}_m$, then $a +_m 0 = a$ and $a \cdot_m 1 = a$.

Arithmetic Modulo m

- *Additive inverses:* If $a \neq 0$ belongs to $\mathbf{Z}_m$, then $m - a$ is the additive inverse of a modulo m and 0 is its own additive inverse.
 - $a +_m (m - a) = 0$ and $0 +_m 0 = 0$
- *Distributivity:* If a , b , and c belong to $\mathbf{Z}_m$, then
 - $a \cdot_m (b +_m c) = (a \cdot_m b) +_m (a \cdot_m c)$ and $(a +_m b) \cdot_m c = (a \cdot_m c) +_m (b \cdot_m c)$.

Multiplicative inverses have not been included since they do not always exist. For example, there is no multiplicative inverse of 2 modulo 6.

(optional) Using the terminology of abstract algebra, $\mathbf{Z}_m$ with $+_m$ is a commutative group and $\mathbf{Z}_m$ with $+_m$ and $\cdot_m$ is a commutative ring.



4.2 Integer Representations and Algorithms

Integer Representations

- Base b Expansions
 - Binary Expansion ($b=2$)
 - Octal Expansion ($b=8$)
 - Hexadecimal Expansion ($b=16$)

Base Conversion Algorithm

Algorithms for Integer Operations



4.2 Integers and Algorithms

Representations of integers

base b expansion of n

$$n = a_k \cdot b^k + a_{k-1} \cdot b^{k-1} + \dots + a_1 \cdot b^1 + a_0 \cdot b^0$$

Binary expansion

Example 1

$$(101011111)_2 = 1 \cdot 2^8 + 0 \cdot 2^7 + 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = (351)_{10}$$

Octal expansion

Example 2

$$(7016)_8 = 7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8^1 + 6 = (3598)_{10}$$

Hexadecimal expansion

Example 3

$$(2AE0B)_{16} = 2 \cdot 16^4 + 10 \cdot 16^3 + 14 \cdot 16^2 + 0 \cdot 16 + 11 = (175627)_{10}$$

4.2 Integers and Algorithms

Base conversion

$$n = bq_0 + a_0 \quad 0 \leq a_0 < b$$

$$q_0 = bq_1 + a_1 \quad 0 \leq a_1 < b$$

$\vdots$

Example 4. Find the base 8, or octal, expansion of $(12345)_{10}$

Sol. $12345 = 8 \cdot 1543 + 1.$

$$1543 = 8 \cdot 192 + 7,$$

$$192 = 8 \cdot 24 + 0,$$

$$24 = 8 \cdot 3 + 0,$$

$$3 = 8 \cdot 0 + 3.$$

$$(12345)_{10} = (30071)_8$$

The remainders are the digits of the base b expansion

4.2 Integers and Algorithms

Base conversion

$$n = bq_0 + a_0 \quad 0 \leq a_0 < b$$

$$q_0 = bq_1 + a_1 \quad 0 \leq a_1 < b$$

$\vdots$

Example 5. Find the hexadecimal expansion of $(177130)_{10}$

Sol. $177130 = 16 \cdot 11070 + 10.$

$$11070 = 16 \cdot 691 + 14,$$

$$691 = 16 \cdot 43 + 3,$$

$$43 = 16 \cdot 2 + 11,$$

$$2 = 16 \cdot 0 + 2. \quad (177130)_{10} = (2B3EA)_{16}$$

The remainders are the digits of the base b expansion

4.2 Integers and Algorithms

Base conversion

$$n = bq_0 + a_0 \quad 0 \leq a_0 < b$$

$$q_0 = bq_1 + a_1 \quad 0 \leq a_1 < b$$

$\vdots$

Example 6. Find the binary expansion of $(241)_{10}$

Sol. $241 = 2 \cdot 120 + 1,$

$$120 = 2 \cdot 60 + 0, \quad 7 = 2 \cdot 3 + 1,$$

$$60 = 2 \cdot 30 + 0, \quad 3 = 2 \cdot 1 + 1,$$

$$30 = 2 \cdot 15 + 0, \quad 1 = 2 \cdot 0 + 1.$$

$$15 = 2 \cdot 7 + 1, \quad (241)_{10} = (11110001)_2 = (F1)_{16}$$

The remainders are the digits of the base b expansion

TABLE 1 (4.2)

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TABLE 1 Hexadecimal, Octal, and Binary Representation of the Integers 0 through 15.																
Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Example 7

$$(11 \underline{111} \underline{010} \underline{111} \underline{100})_2 = (37274)_8 = (3EBC)_{16}$$

$$(765)_8 = (11 \underline{111} \underline{010} \underline{1})_2$$

$$(A8D)_{16} = (101010001101)_2$$

TABLE 1 (4.2)

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Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
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Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Example 7

$$(11 \underline{1110} \underline{1011} \underline{1100})_2 = (37274)_8 = (3EBC)_{16}$$

$$(765)_8 = (111110101)_2$$

$$(A8D)_{16} = (\underline{10101} \underline{0001} \underline{101})_2$$

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Example 8.

$$\begin{array}{rcccc}
 c_3 & c_2 & c_1 & c_0 & \\
 1 & 1 & 1 & 0 & \\
 & 1 & 1 & 1 & 0 \\
 & 1 & 0 & 1 & 1 \\
 \hline
 1 & 1 & 0 & 0 & 1 \\
 s_4 & s_3 & s_2 & s_1 & s_0
 \end{array}$$

$$a_0 + b_0 = c_0 \bullet 2 + s_0,$$

$$a_1 + b_1 + c_0 = c_1 \bullet 2 + s_1,$$

$$a_2 + b_2 + c_1 = c_2 \bullet 2 + s_2,$$

⋮

FIGURE 1 Adding $(1110)_2$ and $(1011)_2$.

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Example 10.

$$\begin{array}{r}
 a: \quad 1 \quad 1 \quad 0 \\
 b: \quad \overset{b_2}{1} \quad \overset{b_1}{0} \quad \overset{b_0}{1} \\
 \hline
 1 \quad 1 \quad 0
 \end{array}$$

$$\begin{array}{r}
 0 \quad 0 \quad 0 \\
 1 \quad 1 \quad 0 \\
 1 \quad 1 \quad 1 \quad 1 \quad 0
 \end{array}$$

$$\begin{aligned}
 ab &= a(b_0 2^0 + b_1 2^1 + \cdots + b_{n-1} 2^{n-1}) \\
 &= a(b_0 2^0) + a(b_1 2^1) + \cdots + a(b_{n-1} 2^{n-1})
 \end{aligned}$$

$$ab_0 \bullet 2^0 = (110)_2 \bullet 1 \bullet 2^0 = (110)_2$$

$$ab_1 \bullet 2^1 = (110)_2 \bullet 0 \bullet 2^1 = (0000)_2$$

$$ab_2 \bullet 2^2 = (110)_2 \bullet 1 \bullet 2^2 = (11000)_2$$

FIGURE 2 Multiplying $(110)_2$ and $(101)_2$.

- 
- **Example 11** : How many additions of bits and shifts of bits are used to multiple a and b with n bits (or less) in their binary representations by Alg. 3?

- 解：在演算法中，使用將部份積 $c_0, c_1, c_2, \dots$ 和 c_{n-1} 相加來求出 a 與 b 的乘積。當 $b_j = 1$ 時，計算部份積 c_j 是將 a 的二進位表示法位移 j 個位置；當 $b_j = 0$ 時，不需要做位移，因為 $c_j = 0$ 。

所以，找出 n 個整數， $ab_j 2^j$ ， $n = 0, 1, 2, \dots, n-1$ ，需要至多

$$0 + 1 + 2 + \dots + n - 1$$

個位移。移動次數為 $O(n^2)$ 。

將整數 ab_j 由 $j = 0$ 加到 $j = n - 1$ ，需要將 n -bit整數加上一個 $(n+1)$ -bit整數， $\dots$ ，加上一個 $(2n)$ -bit整數。根據上面的範例知道每次相加需要做 $O(n)$ 次的位元加法。所以，總共需要的加法次數為 $O(n^2)$ 。

$$n = a_k \cdot b^k + a_{k-1} \cdot b^{k-1} + \dots + a_1 \cdot b^1 + a_0 \cdot b^0$$

Modular Exponentiation

ALGORITHM 5 Modular Exponentiation.

procedure modular exponentiation(b : integer, $n = (a_{k-1}a_{k-2} \dots a_1a_0)_2$,
 m : positive integers)

$x := 1$

$power := b \bmod m$

for $i := 0$ to $k - 1$

 if $a_i = 1$ then $x := (x \cdot power) \bmod m$

$power := (power \cdot power) \bmod m$

return x { x equals $b^n \bmod m$ }

$$3^{644} \bmod 645 = ?$$

$$(644)_{10} = (1010000100)_2 = a_9 \cdot 2^9 + a_7 \cdot 2^7 + a_2 \cdot 2^2$$

$$3^{644} = 3^{(1010000100)_2} = 3^{2^9} \cdot 3^{2^7} \cdot 3^{2^2}$$

Example 12. Use Algorithm 5 to find $3^{644} \bmod 645$.

$i = 0$: Because $a_0 = 0$, we have $x = 1$ and $power = 3^2 \bmod 645 = 9 \bmod 645 = 9$;

$i = 1$: Because $a_1 = 0$, we have $x = 1$ and $power = 9^2 \bmod 645 = 81 \bmod 645 = 81$;

$i = 2$: Because $a_2 = 1$, we have $x = 1 \cdot 81 \bmod 645 = 81$ and $power = 81^2 \bmod 645 = 6561 \bmod 645 = 111$;

$i = 3$: Because $a_3 = 0$, we have $x = 81$ and $power = 111^2 \bmod 645 = 12,321 \bmod 645 = 66$;

$i = 4$: Because $a_4 = 0$, we have $x = 81$ and $power = 66^2 \bmod 645 = 4356 \bmod 645 = 486$;

$i = 5$: Because $a_5 = 0$, we have $x = 81$ and $power = 486^2 \bmod 645 = 236,196 \bmod 645 = 126$;

$i = 6$: Because $a_6 = 0$, we have $x = 81$ and $power = 126^2 \bmod 645 = 15,876 \bmod 645 = 396$;

$i = 7$: Because $a_7 = 1$, we find that $x = (81 \cdot 396) \bmod 645 = 471$ and $power = 396^2 \bmod 645 = 156,816 \bmod 645 = 81$;

$i = 8$: Because $a_8 = 0$, we have $x = 471$ and $power = 81^2 \bmod 645 = 6561 \bmod 645 = 111$;

$i = 9$: Because $a_9 = 1$, we find that $x = (471 \cdot 111) \bmod 645 = 36$.

$$3^{644} \bmod 645 = ?$$

$$3^{644} = 3^{(1010000100)_2} = 3^{2^9} \cdot 3^{2^7} \cdot 3^{2^2}$$

 $a_i,$
 $b=3,$
 $x=\text{remainder},$

$a_0 = 0$

$a_1 = 0$

$a_2 = 1$

$a_3 = 0$

$a_4 = 0$

$a_5 = 0$

$a_6 = 0$

$a_7 = 1$

$a_8 = 0$

$a_9 = 1$

3^{2^0}

3^{2^1}

3^{2^2}

3^{2^3}

3^{2^4}

3^{2^5}

3^{2^6}

3^{2^7}

3^{2^8}

3^{2^9}

ALGORITHM 5 Modular Exponentiation.

```

procedure modular_exponentiation( $b$ : integer,  $n = (a_{k-1}a_{k-2} \dots a_1a_0)_2$ ,
     $m$ : positive integers)
 $x := 1$ 
 $power := b \bmod m$ 
for  $i := 0$  to  $k - 1$ 
    if  $a_i = 1$  then  $x := (x \cdot power) \bmod m$ 
     $power := (power \cdot power) \bmod m$ 
return  $x$  { $x$  equals  $b^n \bmod m$ }
    
```

Example 12. Use Algorithm 5 to find $3^{644} \bmod 645$.

```

 $i = 0$ : Because  $a_0 = 0$ , we have  $x = 1$  and  $power = 3^2 \bmod 645 = 9 \bmod 645 = 9$ ;
 $i = 1$ : Because  $a_1 = 0$ , we have  $x = 1$  and  $power = 9^2 \bmod 645 = 81 \bmod 645 = 81$ ;
 $i = 2$ : Because  $a_2 = 1$ , we have  $x = 1 \cdot 81 \bmod 645 = 81$  and  $power = 81^2 \bmod 645 = 6561 \bmod 645 = 111$ ;
 $i = 3$ : Because  $a_3 = 0$ , we have  $x = 81$  and  $power = 111^2 \bmod 645 = 12,321 \bmod 645 = 66$ ;
 $i = 4$ : Because  $a_4 = 0$ , we have  $x = 81$  and  $power = 66^2 \bmod 645 = 4356 \bmod 645 = 486$ ;
 $i = 5$ : Because  $a_5 = 0$ , we have  $x = 81$  and  $power = 486^2 \bmod 645 = 236,196 \bmod 645 = 126$ ;
 $i = 6$ : Because  $a_6 = 0$ , we have  $x = 81$  and  $power = 126^2 \bmod 645 = 15,876 \bmod 645 = 396$ ;
 $i = 7$ : Because  $a_7 = 1$ , we find that  $x = (81 \cdot 396) \bmod 645 = 471$  and  $power = 396^2 \bmod 645 = 156,816 \bmod 645 = 81$ ;
 $i = 8$ : Because  $a_8 = 0$ , we have  $x = 471$  and  $power = 81^2 \bmod 645 = 6561 \bmod 645 = 111$ ;
 $i = 9$ : Because  $a_9 = 1$ , we find that  $x = (471 \cdot 111) \bmod 645 = 36$ .
    
```



4.3 Primes and Greatest Common Divisors

Def 1. $p \in \mathbb{Z}^+ - \{1\}$ is called **prime** (質數)
if $a \nmid p, \forall 1 < a < p, a \in \mathbb{Z}^+$.

p is called **composite** (合成數) otherwise.

Thm 1. (The fundamental theorem of arithmetic)

Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of nondecreasing size.



4.3 Primes and Greatest Common Divisors

Example 2.

The **prime factorization** (質因數分解) of

$$100 = 2^2 \times 5^2,$$

$$641 = 641,$$

$$999 = 3^3 \times 37,$$

$$1024 = 2^{10}.$$

- 
- **Thm 2 : If n is a composite integer, then n has a primer divisor less than or equal to $\sqrt{n}$**
 - **證明：**若 n 為合成數，則根據合成數的定義，存在因數 a ， $1 < a < n$ ，使得 $a \mid n$ 。故， $n = ab$ ，其中 b 為一個大於1的正整數。若 $a > (n)^{1/2}$ 且 $b > (n)^{1/2}$ ，則 $ab > (n)^{1/2} \cdot (n)^{1/2} = n$ ，會產生矛盾。所以得證 $a \leq (n)^{1/2}$ 且 $b \leq (n)^{1/2}$ 。

■ **Example 3.** Show that 101 is prime

- 解： $\{x \mid x \leq \sqrt{101}, x \in \mathbb{Z}^+\} = \{2, 3, 5, 7\}$
no prime factor not exceeding is found, then n is prime

不大於 $(101)^{1/2}$ 的質數只有2，3，5和7。

因為101都不能被這幾個質數整除，所以101為質數。

TABLE 1 The Sieve of Eratosthenes.

Integers divisible by 2 other than 2 receive an underline.

1	2	3	<u>4</u>	5	<u>6</u>	7	<u>8</u>	9	<u>10</u>
11	<u>12</u>	13	<u>14</u>	15	<u>16</u>	17	<u>18</u>	19	<u>20</u>
21	<u>22</u>	23	<u>24</u>	25	<u>26</u>	27	<u>28</u>	29	<u>30</u>
31	<u>32</u>	33	<u>34</u>	35	<u>36</u>	37	<u>38</u>	39	<u>40</u>
41	<u>42</u>	43	<u>44</u>	45	<u>46</u>	47	<u>48</u>	49	<u>50</u>
51	<u>52</u>	53	<u>54</u>	55	<u>56</u>	57	<u>58</u>	59	<u>60</u>
61	<u>62</u>	63	<u>64</u>	65	<u>66</u>	67	<u>68</u>	69	<u>70</u>
71	<u>72</u>	73	<u>74</u>	75	<u>76</u>	77	<u>78</u>	79	<u>80</u>
81	<u>82</u>	83	<u>84</u>	85	<u>86</u>	87	<u>88</u>	89	<u>90</u>
91	<u>92</u>	93	<u>94</u>	95	<u>96</u>	97	<u>98</u>	99	<u>100</u>

Integers divisible by 3 other than 3 receive an underline.

1	2	3	<u>4</u>	5	<u>6</u>	7	8	<u>9</u>	<u>10</u>
11	<u>12</u>	13	<u>14</u>	<u>15</u>	<u>16</u>	17	<u>18</u>	19	<u>20</u>
<u>21</u>	<u>22</u>	23	<u>24</u>	25	<u>26</u>	<u>27</u>	<u>28</u>	29	<u>30</u>
31	<u>32</u>	<u>33</u>	<u>34</u>	35	<u>36</u>	37	<u>38</u>	<u>39</u>	<u>40</u>
41	<u>42</u>	43	<u>44</u>	<u>45</u>	<u>46</u>	47	<u>48</u>	49	<u>50</u>
<u>51</u>	<u>52</u>	53	<u>54</u>	55	<u>56</u>	<u>57</u>	<u>58</u>	59	<u>60</u>
61	<u>62</u>	<u>63</u>	<u>64</u>	65	<u>66</u>	67	<u>68</u>	<u>69</u>	<u>70</u>
71	<u>72</u>	73	<u>74</u>	<u>75</u>	<u>76</u>	77	<u>78</u>	79	<u>80</u>
<u>81</u>	<u>82</u>	83	<u>84</u>	85	<u>86</u>	<u>87</u>	<u>88</u>	89	<u>90</u>
91	<u>92</u>	<u>93</u>	<u>94</u>	95	<u>96</u>	97	<u>98</u>	<u>99</u>	<u>100</u>

Integers divisible by 5 other than 5 receive an underline.

1	2	3	<u>4</u>	5	<u>6</u>	7	<u>8</u>	<u>9</u>	<u>10</u>
11	<u>12</u>	13	<u>14</u>	<u>15</u>	<u>16</u>	17	<u>18</u>	19	<u>20</u>
<u>21</u>	<u>22</u>	23	<u>24</u>	<u>25</u>	<u>26</u>	27	<u>28</u>	29	<u>30</u>
31	<u>32</u>	<u>33</u>	<u>34</u>	<u>35</u>	<u>36</u>	37	<u>38</u>	<u>39</u>	<u>40</u>
41	<u>42</u>	43	<u>44</u>	<u>45</u>	<u>46</u>	47	<u>48</u>	49	<u>50</u>
<u>51</u>	<u>52</u>	53	<u>54</u>	<u>55</u>	<u>56</u>	<u>57</u>	<u>58</u>	59	<u>60</u>
61	<u>62</u>	<u>63</u>	<u>64</u>	<u>65</u>	<u>66</u>	67	<u>68</u>	<u>69</u>	<u>70</u>
71	<u>72</u>	73	<u>74</u>	<u>75</u>	<u>76</u>	77	<u>78</u>	79	<u>80</u>
<u>81</u>	<u>82</u>	83	<u>84</u>	<u>85</u>	<u>86</u>	<u>87</u>	<u>88</u>	89	<u>90</u>
91	<u>92</u>	<u>93</u>	<u>94</u>	<u>95</u>	<u>96</u>	97	<u>98</u>	<u>99</u>	<u>100</u>

Integers divisible by 7 other than 7 receive an underline; integers in color are prime.

1	2	3	4	5	<u>6</u>	7	8	<u>9</u>	<u>10</u>
11	<u>12</u>	13	<u>14</u>	<u>15</u>	<u>16</u>	17	<u>18</u>	19	<u>20</u>
<u>21</u>	<u>22</u>	23	<u>24</u>	25	26	27	<u>28</u>	29	<u>30</u>
31	<u>32</u>	<u>33</u>	<u>34</u>	<u>35</u>	<u>36</u>	37	<u>38</u>	<u>39</u>	<u>40</u>
41	<u>42</u>	43	<u>44</u>	<u>45</u>	<u>46</u>	47	<u>48</u>	<u>49</u>	<u>50</u>
51	52	53	<u>54</u>	55	<u>56</u>	57	<u>58</u>	59	<u>60</u>
61	<u>62</u>	<u>63</u>	<u>64</u>	65	<u>66</u>	67	<u>68</u>	<u>69</u>	<u>70</u>
71	<u>72</u>	73	<u>74</u>	<u>75</u>	<u>76</u>	77	<u>78</u>	79	<u>80</u>
81	<u>82</u>	83	<u>84</u>	85	<u>86</u>	<u>87</u>	<u>88</u>	89	<u>90</u>
91	<u>92</u>	<u>93</u>	<u>94</u>	95	<u>96</u>	97	<u>98</u>	<u>99</u>	<u>100</u>

Infinitude of Primes



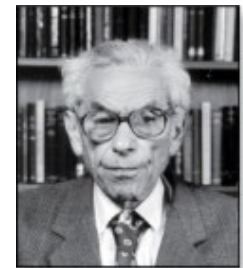
Euclid

Theorem 3: There are infinitely many primes. (Euclid)

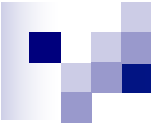
Proof: Assume finitely many primes: $p_1, p_2, \dots, p_n$ (325 B.C.E. – 265 B.C.E.)

- Let $q = p_1 p_2 \cdots p_n + 1$
- Either q is prime or by the fundamental theorem of arithmetic it is a product of primes.
 - But none of the primes p_j divides q since if $p_j \mid q$, then p_j divides $q - p_1 p_2 \cdots p_n = 1$.
 - Hence, there is a prime not on the list $p_1, p_2, \dots, p_n$. It is either q , or if q is composite, it is a prime factor of q . This contradicts the assumption that $p_1, p_2, \dots, p_n$ are all the primes.
- Consequently, there are infinitely many primes.

This proof was given by Euclid *The Elements*. The proof is considered to be one of the most beautiful in all mathematics. It is the first proof in *The Book*, inspired by the famous mathematician Paul Erdős' imagined collection of perfect proofs maintained by God.



Paul Erdős
(1913-1996)

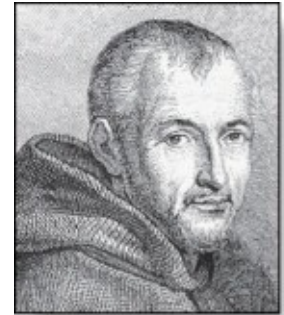


■ Thm 3. 質數有無限多個。

- 證明：我們將使用歸謬證法。假設質數只有有限個， $p_1, p_2, \dots, p_n$ 。令 $Q = p_1 \cdot p_2 \cdot \dots \cdot p_n + 1$ 。

根據算術基本定理， Q 為質數，或是表成兩個或兩個以上質因數的乘積。然而，沒有任何一個質數 p_j 能整除 Q ，由於 $Q - p_1 \cdot p_2 \cdot \dots \cdot p_n = 1$ 。因為沒有一個質數能整除 Q ，所以 Q 本身為質數。如此一來，便與一開始的假設質數矛盾。在這個證明中，我們明確的找出了一個質數不在原來所列之質數序列之中。

Representing Functions



Marin Mersenne

(1588-1648)

Definition: Prime numbers of the form $2^p - 1$, where p is prime, are called *Mersenne primes*.

- $2^2 - 1 = 3$, $2^3 - 1 = 7$, $2^5 - 1 = 31$, and $2^7 - 1 = 127$ are Mersenne primes.
- $2^{11} - 1 = 2047$ is not a Mersenne prime since $2047 = 23 \cdot 89$.
- There is an efficient test for determining if $2^p - 1$ is prime.
- The largest known prime numbers are Mersenne primes.
- As of mid 2011, 47 Mersenne primes were known, the largest is $2^{43,112,609} - 1$, which has nearly 13 million decimal digits.
- The *Great Internet Mersenne Prime Search (GIMPS)* is a distributed computing project to search for new Mersenne Primes.

<http://www.mersenne.org/>

- 
- 近300年來，所能找出的最大質數，都有個特殊的形式， $2^p - 1$ ，其中 p 為質數。這樣的質數稱為梅遜質數(**Mersenne primes**)，根據法國修道士馬梅遜之姓氏命名。之所以知道質數通常有這樣的形式，是因為有個測試法，稱為盧卡司－里莫測試法(**Lucas-Lehmer test**)，能非常有效的判斷出 $2^p - 1$ 的數是否為質數。而且，現今任何測試其他形式的數是否為質數的方法，都沒能如盧卡司－里莫測試法來得有效率。



The distribution of primes

- **質數定理(The Prime Number Theorem)** 當 x 區近於無窮大時，不大於 x 之質數的個數和 $x/\ln x$ 的比值會趨近1。（其中 $\ln x$ 為 x 之自然對數。）



有關質數之猜想與未定論問題

- 哥德巴赫猜想(**Goldbach's Conjecture**) 1742年時，哥德巴赫寫了一封信給尤拉，猜測每一個大於**5**的奇數都能表示成三個質數的和。尤拉回信說，這個猜想其實等價於所有大於**2**的偶數，都能寫成兩個質數之和。現在，這個猜想被稱為哥德巴赫猜想。我們能檢驗幾個較小的偶數： $4 = 2 + 2$ ， $6 = 3 + 3$ ， $8 = 3 + 5$ ， $10 = 3 + 7$ ， $12 = 5 + 7$ 等等。利用現代電腦，哥德巴赫猜想能檢驗至非常大的數。至**2006**年初，這個猜想已經被檢驗至 1.6×10^{18} 。



有關質數之猜想與未定論問題

- **孿生質數猜想(The Twin Prime Conjecture)** 所謂孿生質數指的是一對相差2的質數，如3與5，5與7，11與13，17與19還有4967和4969。孿生質數猜想聲稱這樣的孿生質數有無限多對。目前所知，最接近的證明為：存在無限多對的 p 與 $p + 2$ ，其中 p 為質數，而 $p + 2$ 為質數或是兩質數之積。（於1966年，陳潤證出。）截至2011年初，所知最大的孿生質數為 $65,516,468,355 \times 2^{333,333} \pm 1$ 這兩個數包含100,355位數。



Prime factorization (質因數分解) of

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$$
$$b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n}$$

Def 2. gcd (greatest common divisor)

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)}$$

Def 3. relatively prime (互質)

$$\gcd(a, b) = 1$$

Def 5. lcm (least common multiple)

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)}$$



Example 4.

The prime factorization (質因數分解) of

$$7007 = 7^2 \times 11^1 \times 13^1.$$



Example 10. What is the greatest common divisor of 24 and 36?

The **prime factorization** (質因數分解) 24 and 36

$$24 = 2^3 \times 3^1$$

$$36 = 2^2 \times 3^2$$

$$\gcd(24, 36) = 2^{\min(3, 2)} \times 3^{\min(1, 2)} = 2^2 \times 3^1 = 12$$



Example 11. What is the greatest common divisor of 17 and 22?

$$\gcd(17, 22) = 1$$

Example 12. the integers 17 and 22 are relative prime

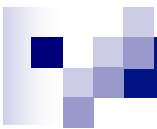
Example 13. Determine whether 10, 17, and 21 are pairwise relative prime and whether the integers 10, 19, and 24 are pairwise relative prime.



Example 15. What is the least common multiple of

$2^3 \times 3^5 \times 7^2$ and $2^4 \times 3^3$?

$$\text{lcm}(2^3 \times 3^5 \times 7^2, 2^4 \times 3^3) = 2^{\max(3,4)} \times 3^{\max(5,3)} \times 7^{\max(2,0)} = 2^4 \times 3^5 \times 7^2$$

- 
- **Exercise 22** : Find $\gcd(1000, 625)$ and $\text{lcm}(1000, 625)$ and verify that $\gcd(1000, 625) \cdot \text{lcm}(1000, 625) = 1000 \cdot 625$

- Sol.

$$\begin{aligned}\gcd(1000, 625) \cdot \text{lcm}(1000, 625) &= \\ 2^{\min(3,0)} \times 5^{\min(3,4)} \cdot 2^{\max(3,0)} \times 5^{\max(3,4)} &= \\ 2^0 \times 5^3 \cdot 2^3 \times 5^4 &= 2^3 \times 5^3 \cdot 2^0 \times 5^4 = \\ 1000 \cdot 625\end{aligned}$$



Ex. How many zeros are there at the end of $100!$?

Sol : 計算 $1 \times 2 \times 3 \times \dots \times 100 = 10^k \times m$, where $10 \nmid m$

$\because 10 = 2 \times 5$, 又2的次數必定比5多

$\therefore$ 計算 $1 \times 2 \times 3 \times \dots \times 100 = 5^k \times n$, where $5 \nmid n$

$\because 5, 10, 15, 20, \dots, 100$ 才有因數5,
而 $25, 50, 75, 100$ 有因數25

$\therefore k=24 \Rightarrow$ 共有24個0



※The Euclidean Algorithm (輾轉相除法求 gcd)

Example : Find $\gcd(91, 287)$

Sol:

$$287 = 91 \cdot 3 + 14 \quad \begin{array}{l} \text{if } x|91 \text{ \& } x|287 \Rightarrow x|14 \\ \therefore \gcd(91, 287) = \gcd(91, 14) \end{array}$$

$$91 = 14 \cdot 6 + 7 \longrightarrow \gcd(91, 14) = \gcd(14, 7)$$

$$14 = 7 \cdot 2 \longrightarrow \gcd(14, 7) = 7$$

$$\therefore \gcd(91, 287) = 7$$

Lemma 1 Let $a = bq + r$, where a, b, q , and $r \in \mathbf{Z}$.
Then $\gcd(a, b) = \gcd(b, r)$.

Euclidean Algorithm

The Euclidian algorithm is an efficient method for computing the greatest common divisor of two integers. It is based on the idea that $\gcd(a,b)$ is equal to $\gcd(a,c)$ when $a > b$ and c is the remainder when a is divided by b .



Euclid

(325 B.C.E. – 265 B.C.E.)

Example: Find $\gcd(91, 287)$:

- $287 = 91 \cdot 3 + 14$ Divide 287 by 91
- $91 = 14 \cdot 6 + 7$ Divide 91 by 14
- $14 = 7 \cdot 2 + 0$ Divide 14 by 7
Stopping condition

$$\gcd(287, 91) = \gcd(91, 14) = \gcd(14, 7) = 7$$



Algorithm 1. (The Euclidean Algorithm)

procedure gcd (a, b : positive integers)

$x := a$

$y := b$

while $y \neq 0$

begin

$r := x \bmod y$ (if $y > x$ then $r = x$)

$x := y$

$y := r$

end

{ gcd (a, b) = x }

eg. 求 gcd (6,12)

$x = 6$

$y = 12$

while $y \neq 0$

$r = 6 \bmod 12 = 6$

$x = 12$

$y = 6$

while $y \neq 0$

$r = 12 \bmod 6 = 0$

$x = 6$

$y = 0$

while $y = 0$, end.

$\therefore \text{gcd} (6,12) = 6$

Exercise : 23,25

- 
- **Example 16** : Find the gcd of 414 and 662 using the Euclidean algorithm

- 解：持續利用除法定理：

$$662 = 414 \cdot 1 + 248$$

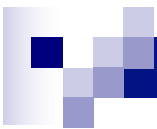
$$414 = 248 \cdot 1 + 166$$

$$248 = 166 \cdot 1 + 82$$

$$166 = 82 \cdot 2 + 2$$

$$82 = 2 \cdot 41$$

所以， $\gcd(414, 662) = 2$ ，因為2是最後一個非零的餘數。



$\gcd(a, b)$ can be expressed as a linear combination with integer coefficients of a and b .

Theorem 6. (Bézout theorem)

If a and b are positive integers, then there exist integers s and t such that $\gcd(a, b) = sa + tb$.

The extended Euclidean Algorithm



Example 17 Express $\gcd(252, 198) = 18$ as a linear combination of 252 and 198.

Sol:

$$252 = 1 \cdot 198 + 54$$

$$198 = 3 \cdot 54 + 36$$

$$54 = 1 \cdot 36 + 18$$

$$36 = 2 \cdot 18$$

$$\therefore \gcd(252, 198) = 18$$

$$18 = 54 - 1 \cdot 36$$

$$36 = 198 - 3 \cdot 54$$

$$54 = 252 - 1 \cdot 198$$

$$\Rightarrow 18 = 54 - 1 \cdot 36 = 54 - 1 \cdot (198 - 3 \cdot 54)$$

$$= 4 \cdot 54 - 1 \cdot 198 = 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198$$

$$= 4 \cdot 252 - 5 \cdot 198$$



Lemma 2.

If a , b and c are positive integers such that $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

Lemma 3.

If p is a prime and $p \mid a_1 a_2 \dots a_n$, where each a_i is an integer, then $p \mid a_i$ for some i .

Example 18

$14 \equiv 8 \pmod{6}$, 但 $\equiv$ 的左右兩邊同除以2後不成立 because $14/2=7$, $8/2=4$, but $7 \not\equiv 4 \pmod{6}$.

另, $14 \equiv 8 \pmod{3}$, 同除以2後, $7 \equiv 4 \pmod{3}$ 成立

Theorem 7.

Let m be a positive integer and let a , b , and c be integers. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

4.4 Linear Congruences (線性同餘)

A congruence (同餘式) of the form $ax \equiv b \pmod{m}$, where m is a positive integer, a and b are integers, and x is a variable, is called a **linear congruence**.

How can we solve the linear congruence $ax \equiv b \pmod{m}$?

Def: If $ax \equiv 1 \pmod{m}$, and let $\bar{a}$ be an answer of x , $\bar{a}$ is called an **inverse** (反元素) of a modulo m



Theorem 1.

If a and m are relatively prime integers and $m > 1$, then an inverse of a modulo m exists.

Furthermore, this inverse is unique modulo m .

Proof. (existence) (unique的部分是exercise)

By Thm 1, because $\gcd(a, m) = 1$,
there exist integers s and t such that $sa + tm = 1$.

$$\Rightarrow sa + tm \equiv 1 \pmod{m}.$$

Because $tm \equiv 0 \pmod{m}$, $sa \equiv 1 \pmod{m}$,

$\Rightarrow s$ is an inverse of a modulo m .

Example 1 Find an inverse of 3 modulo 7.

Sol.

Because $\gcd(3, 7) = 1$, find s, t such that $3s + 7t = 1$.

$$\Rightarrow 7 = 2 \cdot 3 + 1$$

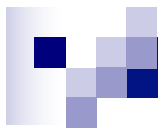
$$\Rightarrow 3s + 7t \equiv 1 \pmod{7}$$

$$\Rightarrow 1 = -2 \cdot 3 + 1 \cdot 7$$

$$\Rightarrow 3s \equiv 1 \pmod{7}$$

$\Rightarrow -2$ is an inverse of 3 modulo 7.

(Note that every integer congruent to -2 modulo 7
is also an inverse of 3, such as 5, -9 , 12, and so on.)



Example: Find an inverse of 101 modulo 4620.

Solution: First use the Euclidian algorithm to show that $\gcd(101, 4620) = 1$.

$$4620 = 45 \cdot 101 + 75$$

$$101 = 1 \cdot 75 + 26$$

$$75 = 2 \cdot 26 + 23$$

$$26 = 1 \cdot 23 + 3$$

$$23 = 7 \cdot 3 + 2$$

$$3 = 1 \cdot 2 + 1$$

$$2 = 2 \cdot 1$$

Since the last nonzero remainder is 1, $\gcd(101, 4260) = 1$

Working Backwards:

$$1 = 3 - 1 \cdot 2$$

$$1 = 3 - 1 \cdot (23 - 7 \cdot 3) = -1 \cdot 23 + 8 \cdot 3$$

$$1 = -1 \cdot 23 + 8 \cdot (26 - 1 \cdot 23) = 8 \cdot 26 - 9 \cdot 23$$

$$1 = 8 \cdot 26 - 9 \cdot (75 - 2 \cdot 26) = 26 \cdot 26 - 9 \cdot 75$$

$$1 = 26 \cdot (101 - 1 \cdot 75) - 9 \cdot 75$$

$$= 26 \cdot 101 - 35 \cdot 75$$

$$1 = 26 \cdot 101 - 35 \cdot (4620 - 45 \cdot 101)$$

$$= -35 \cdot 4620 + 1601 \cdot 101$$

Bézout coefficients : - 35 and 1601

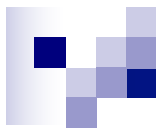
1601 is an inverse of 101 modulo 4620

Inverses in Modular Arithmetic (模算數)

Def: If $a\bar{a} \equiv 1 \pmod{m}$, $\bar{a}$ is called an **inverse** (反元素) of a modulo m , $\bar{a}a - 1 = km$ for some integer k .

we choose as a representative the smallest positive one and set that the inverse $a^{-1} = a \pmod{m}$

- **Example** : Find an inverse of 3 modulo 10.
- 3 has inverse 7 modulo 10 since $3 \cdot (7) \equiv 1 \pmod{10}$ since
$$3 \cdot (7) - 1 = 21 - 1 = 2 \cdot (10)$$



Ex 4 Continuing with example 3 we can write $10 = 5 \cdot 2$. Thus, 3 is relatively prime to 10 and has an inverse modulo 10 while 5 is not relatively prime to 10 and therefore has no inverse modulo 10.

Ex 5 We can compute which numbers will have inverses modulo 10 by computing which are relatively prime to $10 = 5 \cdot 2$. These numbers are $x = 1, 3, 7, 9$. It is easy to see that the following table gives inverses module 10:

Table 1: inverses modulo 10

x	1	3	7	9
$x^{-1} \text{ MOD } 10$	1	7	3	9

Ex 6: We can solve the equation $3 \cdot x + 6 \equiv 8(\text{mod } 10)$ by using the sum (3) and multiplication (4) rules along with the above table:

$$\begin{aligned} 3 \cdot x + 6 &\equiv 8(\text{mod } 10) \implies \\ 3 \cdot x &\equiv 8 - 6 \equiv 2(\text{mod } 10) \implies \\ (3^{-1}) \cdot 3 \cdot x &\equiv (3^{-1}) \cdot 2(\text{mod } 10) \implies \\ x &\equiv 7 \cdot 2(\text{mod } 10) \equiv 14(\text{mod } 10) \equiv 4(\text{mod } 10) \end{aligned}$$

Example 3 What are the solutions of the linear congruence $3x \equiv 4 \pmod{7}$?

Sol.

By **Example 1** $\Rightarrow -2$ is an inverse of 3 modulo 7

$$\Rightarrow 3 \cdot (-2) \equiv 1 \pmod{7}$$

將 $3x \equiv 4 \pmod{7}$ 左右同乘 -2

$$\Rightarrow -2 \cdot 3x \equiv -2 \cdot 4 \pmod{7} \quad \Rightarrow -6x \equiv -8 \pmod{7}$$

Because $-6 \equiv 1 \pmod{7}$, and $-8 \equiv 6 \pmod{7}$,

$\Rightarrow$ If x is a solution, then $x \equiv -8 \equiv 6 \pmod{7}$.

We need to determine whether every x with $x \equiv 6 \pmod{7}$ is a solution. Assume $x \equiv 6 \pmod{7}$, then

$$3x \equiv 3 \cdot 6 = 18 \equiv 4 \pmod{7}.$$

Therefore every such x is a solution:

$$x = 6, 13, 20, \dots, \text{ and } -1, -8, -15, \dots$$

The Chinese Remainder Theorem (中國餘數定理)

Example 4. 孫子算經：「某物不知其數，三三數之餘二，五五數之餘三，七七數之餘二，問物幾何？」(又稱為「韓信點兵」問題)

$$\begin{aligned} \text{i.e.} \quad & x \equiv 2 \pmod{3} \\ & x \equiv 3 \pmod{5} \\ & x \equiv 2 \pmod{7} \end{aligned} \quad x = ?$$

Theorem 2. (The Chinese Remainder Theorem)

Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers and $a_1, a_2, \dots, a_n$ arbitrary integers. Then the system

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$\vdots$$

$$x \equiv a_n \pmod{m_n}$$

has a unique solution modulo $m = m_1 m_2 \dots m_n$.

(即有一解 x , where $0 \leq x < m$, 且所有其他解 mod m 都等於 x)

Proof of Thm 2:

$$m = m_1 m_2 \dots m_n$$

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ &\vdots \\ x &\equiv a_n \pmod{m_n} \end{aligned}$$

$$\text{Let } M_k = m / m_k \quad \forall 1 \leq k \leq n$$

$\because m_1, m_2, \dots, m_n$ are pairwise relatively prime

$$\therefore \gcd(M_k, m_k) = 1$$

$$\Rightarrow \exists \text{ integer } y_k \text{ s.t. } M_k y_k \equiv 1 \pmod{m_k}$$

(by Thm. 3)

$$\Rightarrow a_k M_k y_k \equiv a_k \pmod{m_k}, \quad \forall 1 \leq k \leq n$$

$$\text{Let } x = a_1 M_1 y_1 + a_2 M_2 y_2 + \dots + a_n M_n y_n$$

$$\because m_i \mid M_j, \quad \forall i \neq j$$

$$\therefore x \equiv a_k M_k y_k \equiv a_k \pmod{m_k} \quad \forall 1 \leq k \leq n$$

x is a solution.

All other solution y satisfies $y \equiv x \pmod{m_k}$.

$$\begin{array}{l} x \equiv 2 \pmod{3} \\ x \equiv 3 \pmod{5} \\ x \equiv 2 \pmod{7} \end{array} \quad x = ?$$

Example 5. (Solve the system in Example 5)

$$\text{Let } m = m_1 m_2 m_3 = 3 \cdot 5 \cdot 7 = 105$$

$$M_1 = m / m_1 = 105 / 3 = 35 \text{ (也就是 } m_2 m_3 \text{)}$$

$$M_2 = m / m_2 = 105 / 5 = 21$$

$$M_3 = m / m_3 = 105 / 7 = 15$$

找 y_1 使得 $M_1 y_1 \equiv 1 \pmod{3}$

$$35 \equiv 2 \pmod{3} \Rightarrow \overset{M_1}{35} \times \overset{y_1}{2} \equiv 1 \pmod{3}$$

$$21 \equiv 1 \pmod{5} \Rightarrow \overset{M_2}{21} \times \overset{y_2}{1} \equiv 1 \pmod{5}$$

$$15 \equiv 1 \pmod{7} \Rightarrow \overset{M_3}{15} \times \overset{y_3}{1} \equiv 1 \pmod{7}$$

$$\therefore x = a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3$$

$$= 2 \times 35 \times 2 + 3 \times 21 \times 1 + 2 \times 15 \times 1 = 233 \equiv 23 \pmod{105}$$

$\therefore$ 最小的解為23，其餘解都等於 $23 + 105t$ for some $t \in \mathbf{Z}^+$



Exercise. Find all solutions to the system of congruences

$$x \equiv 2 \pmod{3}$$

$$x \equiv 1 \pmod{4}$$

$$x \equiv 3 \pmod{5}$$

Sol :

$$a_1=2, a_2=1, a_3=3,$$

$$m_1=3, m_2=4, m_3=5 \quad m=3 \times 4 \times 5=60$$

$$M_1=20, M_2=15, M_3=12$$

$$20 \equiv 2 \pmod{3} \Rightarrow 20 \overset{M_1}{\times} \overset{y_1}{2} \equiv 1 \pmod{3}$$

$$15 \equiv 3 \pmod{4} \Rightarrow 15 \overset{M_2}{\times} \overset{y_2}{3} \equiv 1 \pmod{4}$$

$$12 \equiv 2 \pmod{5} \Rightarrow 12 \overset{M_3}{\times} \overset{y_3}{3} \equiv 1 \pmod{5}$$

$$\therefore x = 2 \times 20 \times 2 + 1 \times 15 \times 3 + 3 \times 12 \times 3$$

$$= 80 + 45 + 108 = 233 \equiv 53 \pmod{60}$$



Example 6. To find all integers x such that
 $x \equiv 1(\text{mod } 5)$, $x \equiv 2(\text{mod } 6)$, and $x \equiv 3(\text{mod } 7)$.

■ Use *back substitution*,

$$\begin{aligned}\because x &\equiv 1(\text{mod } 5) \\ x &= 5t + 1\end{aligned}$$

$$\begin{aligned}\because x &\equiv 2(\text{mod } 6) \\ x = 5t + 1 &\equiv 2(\text{mod } 6) \\ t &\equiv 5(\text{mod } 6)\end{aligned}$$

$$x = 5t + 1 = 5(6u + 5) + 1 = 30u + 26$$

$$\begin{aligned}\because 30u + 26 &\equiv 3(\text{mod } 7) \\ u &\equiv 6(\text{mod } 7)\end{aligned}$$

$$x = 30u + 26 = 30(7v + 6) + 26 = 210v + 206$$

Example 6. To find all integers x such that

$$\begin{aligned}x &\equiv 1 \pmod{5} \\x &\equiv 2 \pmod{6} \\x &\equiv 3 \pmod{7}\end{aligned}$$

Use *Chinese remainder theorem*,

Sol : $a_1=1$, $a_2=2$, $a_3=3$,
 $m_1=5$, $m_2=6$, $m_3=7$ $m=5\times 6\times 7=210$
 $M_1=42$, $M_2=35$, $M_3=30$
if $m_1, m_2, \dots, m_n$ are pairwise relatively prime
then $\gcd(M_k, m_k) = 1$
 $\Rightarrow \exists$ integer y_k s.t. $M_k y_k \equiv 1 \pmod{m_k}$

$$M_1 \cdot 42 \times 3^{y_1} \equiv 1 \pmod{5}$$

$$M_2 \cdot 35 \times 5^{y_2} \equiv 1 \pmod{6}$$

$$M_3 \cdot 30 \times 4^{y_3} \equiv 1 \pmod{7}$$

$$\begin{aligned}\therefore x &= 1 \times 42 \times 3 + 2 \times 35 \times 5 + 3 \times 30 \times 4 \\&= 126 + 350 + 360 = 836 \equiv 206 \pmod{210}\end{aligned}$$

$$\therefore 206, 416, 626, 836, \dots$$



Computer Arithmetic with Large Integers

Suppose that $m_1, m_2, \dots, m_n$ be pairwise relatively prime integers greater than or equal to 2 and let $m = m_1 m_2 \dots m_n$.

By the Chinese Remainder Theorem, we can show that an integer a with $0 \leq a < m$ can be uniquely represented by the n -tuple $(a \bmod m_1, a \bmod m_2, \dots, a \bmod m_n)$.

Example 7 What are the pairs used to represent the nonnegative integers $x < 12$ when they are represented by the order pair $(x \bmod 3, x \bmod 4)$?

Sol

$$\begin{array}{llllll} 0=(0, 0), & 1=(1, 1), & 2=(2, 2), & 3=(0, 3), & 4=(1, 0), & 5=(2, 1), \\ 6=(0, 2), & 7=(1, 3), & 8=(2, 0), & 9=(0, 1), & 10=(1, 2), & 11=(2, 3). \end{array}$$



To perform arithmetic with larger integers, we select moduli (modulus的複數) $m_1, m_2, \dots, m_n$, where each m_i is an integer greater than 2, $\gcd(m_i, m_j) = 1$ whenever $i \neq j$, and $m = m_1 m_2 \dots m_n$ is greater than the result of the arithmetic operations we want to carry out.



Example 8 Suppose that performing arithmetic with integers less than 100 on a certain processor is much quicker than doing arithmetic with larger integers. We can restrict almost all our computations to integers less than 100 if we represent integers using their remainders modulo pairwise relatively prime integers less than 100.

For example, 99, 98, 97, and 95 are pairwise relatively prime. every nonnegative integer less than $99 \times 98 \times 97 \times 95 = 89403930$ can be represented uniquely by its remainders when divided by these four moduli.

E.g., $123684 = (33, 8, 9, 89)$, and $413456 = (32, 92, 42, 16)$

$$\begin{aligned} 123684 + 413456 &= (33, 8, 9, 89) + (32, 92, 42, 16) \\ &= (65 \bmod 99, 100 \bmod 98, 51 \bmod 97, 105 \bmod 95) \\ &= (65, 2, 51, 10) \end{aligned}$$

Use Chinese Remainder Thm to find this sum $\Rightarrow 537140$.

Theorem 3 (Fermat's Little Theorem)

If p is prime and a is an integer not divisible by p , then

$$a^{p-1} \equiv 1 \pmod{p}$$

Furthermore, for every integer a we have

$$a^p \equiv a \pmod{p}$$



Pierre de Fermat
(1601-1665)

Exercise 25(a) Show that $2^{340} \equiv 1 \pmod{11}$
by Fermat's Little Theorem and noting that $2^{340} = (2^{10})^{34}$.

Proof 11 is prime and 2 is an integer not divisible by 11.

$$\Rightarrow 2^{10} \equiv 1 \pmod{11}$$

$$\Rightarrow 2^{340} \equiv 1 \pmod{11} \quad \text{by Thm 5 of Sec. 4.1}$$

$$(a \equiv b \pmod{m} \text{ and } c \equiv d \pmod{m}) \\ \Rightarrow ac \equiv bd \pmod{m}$$

Exercise : Compute $5^{2003} \pmod{7}$

Theorem 3 (Fermat's Little Theorem)

If p is prime and a is an integer not *divisible by* p , then

$$a^{p-1} \equiv 1 \pmod{p}$$

Furthermore, for every integer a we have

$$a^p \equiv a \pmod{p}$$

3. Calculate $7^{121} \bmod 13$ by Fermat's Little Theorem

13 is prime and 7 is an integer not divisible by 13.

$$\Rightarrow 7^{12} \equiv 1 \pmod{13}$$

$$\Rightarrow 7^{12 \cdot 10} \equiv 1 \pmod{13} \quad \text{by Thm 5 of Sec. 4.1}$$

$$(a \equiv b \pmod{m} \text{ and } c \equiv d \pmod{m})$$

$$\Rightarrow ac \equiv bd \pmod{m}$$

$$\Rightarrow 7^{121} \bmod 13 \equiv 7^1 \cdot 7^{12 \cdot 10} \bmod 13 \equiv 7^1 \cdot 1 \pmod{13} = 7$$



Pseudoprimes

By Fermat's little theorem $n > 2$ is prime, where

$$2^{n-1} \equiv 1 \pmod{n}.$$

But if this congruence holds, n may not be prime. Composite integers n such that $2^{n-1} \equiv 1 \pmod{n}$ are called *pseudoprimes* to the base 2.

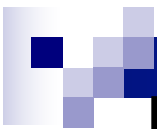
Example: The integer 341 is a pseudoprime to the base 2.

$$341 = 11 \cdot 31$$

$$2^{340} \equiv 1 \pmod{341} \text{ (see in Exercise 37)}$$

We can replace 2 by any integer $b \geq 2$.

Definition: Let b be a positive integer. If n is a composite integer, and $b^{n-1} \equiv 1 \pmod{n}$, then n is called a *pseudoprime to the base b* .



Pseudoprimes

Given a positive integer n , such that $2^{n-1} \equiv 1 \pmod{n}$:

- If n does not satisfy the congruence, it is composite.
- If n does satisfy the congruence, it is either prime or a pseudoprime to the base 2.

Doing similar tests with additional bases b , provides more evidence as to whether n is prime.

Among the positive integers not exceeding a positive real number x , compared to primes, there are relatively few pseudoprimes to the base b .

- For example, among the positive integers less than 10^{10} there are 455,052,512 primes, but only 14,884 pseudoprimes to the base 2.

Carmichael Numbers

(*optional*)

Robert
Carmichael
(1879-1967)



There are composite integers n that pass all tests with bases b such that $\gcd(b, n) = 1$.

Definition: A composite integer n that satisfies the congruence $b^{n-1} \equiv 1 \pmod{n}$ for all positive integers b with $\gcd(b, n) = 1$ is called a *Carmichael* number.

Example: The integer 561 is a Carmichael number. To see this:

- 561 is composite, since $561 = 3 \cdot 11 \cdot 13$.
- If $\gcd(b, 561) = 1$, then $\gcd(b, 3) = 1$, then $\gcd(b, 11) = \gcd(b, 17) = 1$.
- Using Fermat's Little Theorem: $b^2 \equiv 1 \pmod{3}$, $b^{10} \equiv 1 \pmod{11}$, $b^{16} \equiv 1 \pmod{17}$.
- Then

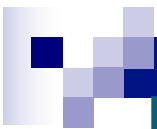
$$b^{560} = (b^2)^{280} \equiv 1 \pmod{3},$$

$$b^{560} = (b^{10})^{56} \equiv 1 \pmod{11},$$

$$b^{560} = (b^{16})^{35} \equiv 1 \pmod{17}.$$

- It follows (see *Exercise 29*) that $b^{560} \equiv 1 \pmod{561}$ for all positive integers b with $\gcd(b, 561) = 1$. Hence, 561 is a Carmichael number.

Even though there are infinitely many Carmichael numbers, there are other tests (described in the exercises) that form the basis for efficient probabilistic primality testing. (see *Chapter 7*)



Primitive Roots

Definition: A primitive root modulo a prime p is an integer r in $\mathbf{Z}_p$ such that every nonzero element of $\mathbf{Z}_p$ is a power of r .

Example: Since every element of $\mathbf{Z}_{11}$ is a power of 2, 2 is a primitive root of 11.

Powers of 2 modulo 11: $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 5$, $2^5 = 10$, $2^6 = 9$, $2^7 = 7$, $2^8 = 3$, $2^{10} = 2$.

Example: Since not all elements of $\mathbf{Z}_{11}$ are powers of 3, 3 is not a primitive root of 11.

Powers of 3 modulo 11: $3^1 = 3$, $3^2 = 9$, $3^3 = 5$, $3^4 = 4$, $3^5 = 1$, and the pattern repeats for higher powers.

Important Fact: There is a primitive root modulo p for every prime number p .

Discrete Logarithms

Suppose p is prime and r is a primitive root modulo p . If a is an integer between 1 and $p-1$, that is an element of $\mathbf{Z}_p$, there is a unique exponent e such that $r^e = a$ in $\mathbf{Z}_p$, that is, $r^e \bmod p = a$.

Definition: Suppose that p is prime, r is a primitive root modulo p , and a is an integer between 1 and $p-1$, inclusive. If $r^e \bmod p = a$ and $1 \leq e \leq p-1$, we say that e is the *discrete logarithm* of a modulo p to the base r and we write $\log_r a = e$ (where the prime p is understood).

Example 1: We write $\log_2 3 = 8$ since the discrete logarithm of 3 modulo 11 to the base 2 is 8 as $2^8 = 3$ modulo 11.

Example 2: We write $\log_2 5 = 4$ since the discrete logarithm of 5 modulo 11 to the base 2 is 4 as $2^4 = 5$ modulo 11.

There is no known polynomial time algorithm for computing the discrete logarithm of a modulo p to the base r (when given the prime p , a root r modulo p , and a positive integer $a \in \mathbf{Z}_p$). The problem plays a role in cryptography as will be discussed in Section 4.6.